



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

PROFESSIONAL SKILL CERTIFICATE

SCHOOL OF PROFESSIONAL & CONTINUING EDUCATION (SPACE)

UNIVERSITI TEKNOLOGI MALAYSIA

GLRB0010 - DESIGN THINKING FOR ENTREPERENEUR

PRE-CLASS ASSIGNMENT

SESSION 2026/2027 SEMESTER 2

PROGRAMME: SECPH

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1. **Design Thinking (DT) is described by Brown in [1] as a process that must pass through three spaces. Then, DT has been evolving and becoming a more refined model as the 5-steps Design Thinking Process shown in the video by InVision. Describe the similarity between the DT process defined by Brown [1] and InVision [2].**

Brown's Design Thinking process is founded on three spaces: inspiration, ideation, and implementation. This is very similar to the InVision video's broad model that also divides the Design Thinking process into three spaces: immersion, ideation and implementation. InVision's 5-step model simply breaks down these spaces:

- "Empathize" and "Define" steps match Brown's Inspiration space.
- The "Ideate" step is Brown's Ideation space.
- The "Prototype" and "Test" steps equate to Brown's Implementation phase.

Moreover, both models stress that these steps are not necessarily linear; they are very iterative, meaning teams will often learn something new that will require them to repeat one or more of these steps. These models are not only similar in structure, but they also share a philosophy that design should be human-centred and iterative. The alternative way from Brown's 3-space model to InVision's 5-step model is not a leap in philosophy but one in practicability. The 5 steps provide more practical advice for each step while adhering to Brown's philosophy.

2. **The step called 'Ideate' [2] is a very important and crucial part in Design Thinking. Elaborate this 'Ideate' step and explain how it should be executed in order to offer effective solutions.**

This is the step where the problem statement progresses into a wide-ranging brainstorm. To carry out this step, teams should focus on generating quantity rather than quality, withhold judgment and allow for "crazy ideas". Innovation is a multi-faceted process, so it is important that this step be conducted by cross-disciplinary teams.

Brainstorming should not be left up to chance, but rather guided through the use of specific ideation techniques (e.g., asking "How Might We" questions, Mind Mapping, or the SCAMPER framework) to prompt the team to think about the problem in different ways. Finally, this long list of ideas needs to be narrowed down (e.g., by dot-voting or feasibility matrices) to identify the most promising solutions to be prototyped.

3. **Describe how the IDEO Design Thinking team has helped the shopping cart problem [3].**

IDEO's solution for the problem of shopping carts was immersion and rapid prototyping. They started by dividing into groups to observe and interview the "experts": consumers, grocery store owners, and cart repairers. This observation brought out extreme but

under-acknowledged problems, such as child safety concerns (causing 22,000 hospitalizations annually), the resulting financial losses from cart theft and physical problems such as an empty plastic cart being blown away.

They then used targeted brainstorming, with "wild ideas" and no judgment. They then constructed four different prototypes within a day, focussed on different aspects (safety, handbasket modularity, high-tech checkout, etc.) and eventually integrated the best, most manufacturable features from each into a new, highly innovative product.

4. Based on the IDEO [3], multi-disciplinary team formation is one of the requirements to ensure successful design. Elaborate these points in the context of design.

Cross-disciplinary team building provides a holistic perspective for a design to be applicable, useful and buildable from different perspectives, avoiding the myopia of a single discipline. The IDEO project team was highly diverse, featuring a project manager selected for his group dynamics skills, rather than his seniority, a Harvard MBA, a linguist, a marketer, a psychologist and a biology major.

When it comes to the Design Thinking process, each discipline has its role. A psychologist and linguist are helpful in the Empathize and Define stages where human behaviour and definition of the user needs are key. A business or marketing specialist is critical during Ideate and Prototype to keep ideas viable. A biology major or technical expert ensures the final prototype is technically feasible.

Without having all these viewpoints at the table, the team runs the risk of designing solutions that are either unbuildable, unsustainable or simply miss the point. In short, this interdisciplinary overlapping demonstrates the IDEO mantra that "enlightened trial and error succeeds over the planning of the lone genius".

5. Share your view on the main difference between the Design Thinking and conventional design approach in designing solutions. Also give your views on the important factors that should be considered in the Design Thinking approach to ensure effective solutions can be proposed.

The key difference is how they begin: traditional design often starts with a linear approach focusing on technical or pre-defined business needs, while Design Thinking focuses on a non-linear approach with strong human empathy to understand the user's real needs and pain points.

To propose successful solutions, four elements need to be present in the Design Thinking process:

- Keeping User Empathy Front of Mind: Business objectives must be kept secondary and the user's needs must be the "North Star".
- Cross-disciplinary Collaboration: Cross-disciplinary teams must be engaged to assess the problem from psychological, engineering and business perspectives to avoid myopia.
- Fail, Fail, Fail: A "Fail often to succeed sooner" approach must be taken. Ideas need to be prototyped quickly and tested by users to get feedback and improve the solution before significant financial commitments are made.
- Continuously Involving the User: The user should not be seen at the Empathize stage and then at the Test stage again. Regular feedback cycles at each iteration can prevent the solution from falling prey to "team bias".